

PSYCHOLOGY, PSYCHIATRY & BRAIN NEUROSCIENCE SECTION

Intensity of Chronic Low Back Pain and Activity Interference: A Daily Diary Study of the Moderating Role of Cognitive Pain Coping Strategies

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Abstract

Objective. Chronic low back pain (CLBP) has a significant negative impact on daily functioning, particularly for those with challenges coping adaptively with ongoing pain. However, the dynamics of pain coping in daily life remain understudied. Therefore, we examined the extent to which pain intensity interferes with daily activities, and assessed whether pain coping strategies (as assessed using daily diaries) moderated this link. **Method.** We analyzed diary data from a sample of 84 participants with CLBP who completed daily diaries for up to 30 days rating pain intensity, pain interference with daily activities, and their use of pain coping strategies, including pain rumination (i.e., repetitive thinking about the pain and its causes), reappraisal (i.e., evaluating one's pain less negatively or more positively), and distraction (i.e., diverting attention from the pain). We hypothesized that these coping strategies would moderate the associations between pain and pain interference with daily activities, although in different directions. **Results.** Results suggest that pain rumination strengthens the association between pain intensity and pain interference both on the person and day level, while pain reappraisal and distraction weaken this association, at the day and person levels, respectively. **Conclusion.** Our findings suggest that those who are more preoccupied with their pain and those who are less likely to reappraise their pain have more pain interference with daily activities. These findings build on prior work on pain coping by using daily diaries and highlight two pain coping strategies that have particular relevance for reducing the impact of CLBP in daily life.

Key Words: Chronic Pain; Daily Diary; Pain Coping Strategies; Pain Interference

Chronic low back pain (CLBP) is one of the most prevalent ailments in the modern world, resulting in major

disability, human suffering, and financial costs [1–3]. CLBP can interfere with individuals' daily physical

functioning and can reduce their quality of life, as it can have significant effects on individuals' ability to perform daily life activities, from basic activities such as dressing and walking to various work-related behaviors [3–5]. However, not all people with CLBP experience significant impairment in their daily functioning. Individual differences in CLBP impairment may be explained, in part, by differences in pain coping [6–9].

Patients with chronic pain can use a variety of strategies to manage their pain [10]. In the current study we focused on coping strategies that lie in the intersection between affective science and pain science. Specifically, we focused on cognitive coping strategies, as they were previously found to be among the most used methods to cope with pain, after pain medication [11, 12]. Thus, we are adopting an affect regulation perspective which prioritizes a set of theoretically important affect regulation processes, namely rumination, reappraisal, and distraction. Pain rumination, a characteristic of pain catastrophizing (which involves an “exaggerated cognitive and affective reaction to an expected or actual pain experience” [6].), entails repetitive thinking about the pain and its causes, is associated with greater disability [13, 14], pain intensity [6] and depression [15]. Pain rumination is generally considered as a maladaptive form of coping strategy.

Pain reappraisal is a more adaptive cognitive pain coping strategy. Pain reappraisal involves finding a more positive meaning to the pain experience or evaluating one's pain less negatively or more positively, and it has been associated with reduced pain intensity [7, 16]. Although it is an effective coping strategy, it is also one of the most cognitively complex coping strategies [7]. Another common way to attempt to control pain is distraction. Pain distraction entails diverting attention from the pain, in an attempt to not feel the pain [8]. However, unlike rumination, which is largely considered a maladaptive pain coping strategy, and reappraisal, which is largely considered an adaptive pain coping strategy, research on pain distraction has yielded mixed results in terms of its impact on pain and pain interference [8, 17, 18].

So far, research on the links among pain intensity, pain interference, and pain coping strategies has been limited. Prior studies have focused on mean levels of pain intensity. However, relying solely on mean levels of pain intensity may not provide a comprehensive understanding on the associations among pain intensity, pain interference, and pain coping, as pain intensity tends to fluctuate daily [19, 20]. Moreover, these studies have mainly relied on surveys that used a timeframe of a week or more [21], which renders findings subject to recall bias.

One promising way to address the shortcomings of previous research is to use daily diaries in which subjects provide daily data about experiences and behaviors in their natural environment [20]. Daily diaries offer several

advantages, including minimizing recall bias and recency effects and peak experiences, thus enhancing data quality and precision [20, 22–25]. As such, daily diaries may be used to clarify within-person processes with greater precision and answer clinically relevant questions that elucidate individual differences in pain responses as a function of pain appraisals [26, 27].

The aim of the current study was to examine the extent to which pain intensity interferes with daily activities and to assess whether pain coping strategies moderated this link. To achieve this, we analyzed daily diaries data from people with CLBP. Over 30 days, participants completed daily diaries to rate their pain; the extent to which pain interfered with their day's activities; and pain rumination, reappraisal, and distraction. We chose these strategies as they are among the most important, studied, and substantial coping strategies [13, 14, 16, 17]. First, based on the literature, we hypothesized that pain intensity would be positively associated with pain interference, both at the person and day levels. Second, we hypothesized that pain rumination would act as a risk factor for pain interference, while reappraisal would act as a resilience factor. Due to the prior mixed findings on distraction, we did not have a directional hypothesis regarding distraction. To our knowledge, this is the first study to utilize daily diaries to examine whether pain coping strategies moderate the daily associations of pain intensity with pain interference in CLBP.

Methods

Study Design and Participants

Participants were 84 people with CLBP (50.6% women) who were part of a larger study of mind-body therapies for CLBP (NCT02503475) and completed the relevant daily diary measures for up to 30 days ($M = 14.57$, $SD = 12.00$), prior to their exposure of the intervention. Overall, 1,562 daily diary entries, which were collected between March 2015 and March 2016, were eligible for data analyses. The Stanford University School of Medicine Institutional Review Board (IRB) approved the protocol used for this study. The data are available upon request.

Inclusion criteria were being between the age of 21 and 65 years; having chronic axial low back pain without radicular symptoms; duration of symptoms greater than three months; past month average pain intensity rating $\geq 4/10$; and English fluency. Chronic low back pain was defined using the National Institutes of Health (NIH) Task Force Standards for Chronic Low Back Pain [28] as pain with a duration greater than 3 months, a frequency of at least half the days in the past 6 months, and localization in the space between below the lower posterior margin of the rib cage and the horizontal gluteal fold.

Participants were recruited from the Stanford Pain Management Center through clinical referral and the San

Francisco Bay Area surrounding community through advertisements. Potential participants contacted the team and were screened through a phone interview for eligibility to participate in the study. Participants completed the diary once a day. These questionnaires were completed online or via an Android device that was supplied to them. Typically, the daily questionnaire took less than 5 minutes to complete.

Measures

Demographic Characteristics

Participants were asked questions regarding their demographic characteristics. Regarding race, participants were asked to choose “which of the following groups do you most strongly consider yourself to be a member of?” Regarding education, participants were asked to indicate the highest grade or level of school completed or the highest degree obtained. Regarding income, participants were asked to indicate their annual household income.

Daily Diaries

The daily diaries included items about pain, mood, and physical functioning; participants were asked to reference the previous 24 hours while considering each item. Each item was rated using a manual slider scale (0–100). Each day participants received an e-mail containing a link to the daily survey. Participants answered the diaries between 5 pm and midnight. See [supplementary materials](#) for an illustration of the daily diary questions used in the current study.

Pain Intensity and Pain Interference

Average pain intensity was assessed with a single item asking participants to rate their “average level of pain” using the following scale anchors: 0 = *Not at all*, to 100 = *Worst pain imaginable*. Activity interference by pain was assessed with the following item and anchors: “How much did low back pain interfere with your daily activities today?” (0 = *Not at all*, to 100 = *Very much*).

Cognitive Pain Coping

Coping regulation strategies were measured by single items, as is typical in most daily diaries studies [29]. Pain rumination was assessed with the following item and anchors: “Today, I couldn’t seem to keep my pain out of my mind” (0 = *Not at all*, to 100 = *All the time* [29]). The validity of the item was assessed using Pearson correlations between the item and the participants’ mean score of the rumination subscale of the Pain Catastrophizing Scale (PCS [30]). The validity coefficient was $r = 0.49$ (95% CI [0.26, 0.67] $P < .001$). Pain reappraisal was assessed with the following item and anchors: “How much did you try to view your pain less negatively today?” (0 = *Not at all*, to 100 = *All the time*; [29, 31]). The validity of the item was assessed using Pearson

correlations between the item and the participants’ mean score of the Emotion Regulation Questionnaire (ERQ[32], that assess respondents’ tendency to regulate their emotions using cognitive reappraisal. The validity coefficient was $r = 0.34$ (95% CI [0.09, 0.55] $P < .01$). Distraction was assessed with the following item and anchors: “How much did you try to distract yourself from your low back pain symptoms today?” (0 = *Not at all*, to 100 = *very much*; [31]). The validity of the item was assessed using Pearson correlations between the item and the participants’ mean score of the Rumination Scale Questionnaire (RSQ [33]), the distraction subscales. The validity coefficient was $r = 0.26$ (95% CI [0.01, 0.49] $P < .05$). The questionnaires used for validity analyses were completed a few weeks after the completion of the diaries’ assessment.

Data Analyses

First, we computed the Pearson correlation between the average levels of pain intensity and pain interference with daily activities and between pain interference and pain coping. Next, we tested for moderation effects, first on the person-level and then on the day-level. The person-level analysis enabled us to test individual differences between participants, while the day-level analysis enabled us to test within-person differences across the days of the study. For each level of analysis, that is, person and day level, we first entered participants’ gender, age, race, education, and income as control variables. However, none of these control variables was significant and therefore were not included in the final models.

To test whether pain coping strategies moderated the associations between pain intensity and pain interference on the person level, we averaged all the variables across each participants’ assessment period, and then standardized the person-level variables (i.e., average pain intensity, pain interference, and pain coping strategies) to make the interpretation of intercepts and slopes clearer (i.e., the estimates reflect effect sizes), and to allow testing of interaction effects. Next, we conducted a moderation analysis using only person-level variables and included all the pain coping strategies in one model.

To test whether pain coping moderated the associations between pain intensity and pain interference on the day level ($M = 14.57$, $SD = 12$), we had to consider the hierarchical nature of the data, that is, days were nested within individuals. To account for this model structure, we used multilevel modeling, and used a compound symmetry structure across the daily errors and included participants’ daily pain intensity as a random slope. All analyses were conducted in R using the nlme package [34]. All variables were standardized.

In the day-level analyses, we entered several covariates. First, we entered the previous day’s pain interference score as a covariate, so that any moderation effect could not be explained by yesterday’s pain interference. Previous day’s

pain and pain coping were not entered as covariates because they were not hypothesized to influence on today's pain, which was supported in the conducted analyses. Second, we entered the person-level pain intensity and pain coping strategies as covariates, which allowed us to estimate both the person-level and day-level effects separately and simultaneously [35]. Adding the person-level effects to the day-level model enabled us to exclude person-dependent "third variables" (e.g., personality, mood) as alternative explanations, as they were embedded in the person-level variance. In all of the analyses we applied a general alpha of 0.05. Finally, we conducted post-hoc analyses using the Johnson-Neyman (J-N) technique [36] on the significant interactions to identify the values of the pain coping strategies for which the slope of the pain predictor was significant.

Results

Demographic characteristics for the sample are presented in Table 1. Not all participants reported on all their demographic characteristics, and so we did not have information on all participants. Participants' mean age was 41.39 (SD = 12.20, range 21– 64). Almost third of the participants identified as Caucasian (29.8%), and 72.3% had a bachelor or advanced degree.

Means and standard deviations at the person level (averaged across the time points for each person) are reported in Table 2, as well as zero-order correlations between the variables. On average, participants reported moderate levels of pain intensity, and indicated that their pain intensity moderately interfered with their daily activities. Regarding associations with pain intensity and

pain coping strategies, pain interference was primarily associated with pain intensity, pain rumination and distraction, and to a lesser degree with pain reappraisal.

Person-Level Moderation

The results of the person-level moderation model are presented in Table 3 and Figure 1. As can be seen, pain interference was significantly and positively related to pain intensity. In addition, rumination and distraction were significant moderators in the associations between person-level pain and pain interference (see Figures 1A and 1B for rumination and distraction, respectively).

The results of the J-N technique indicated that the associations between pain intensity and pain interference increased in tandem with rumination. In comparison, the associations between pain intensity and pain interference decreased with greater use of distraction (see Figures 1A and 1B for rumination and distraction, respectively), to the point where the associations became insignificant for above average levels of distraction (see the purple background above the zero point in Figure 1B).

Day-Level Moderation

The results of the day-level moderation analyses are presented in Table 4. The association between daily pain intensity and pain interference was moderated by daily pain rumination and daily pain reappraisal (see Figure 2A and 2B for daily rumination and reappraisal, respectively). Again, we conducted post-hoc analyses using the J-N technique [36] on the significant interactions (see Figures 2A and 1B for daily rumination and reappraisal, respectively). The results indicated that the associations between pain intensity and pain interference increased the more participants used rumination, while it decreased the more they used reappraisal.

Discussion

To our knowledge, this is the first study to examine the moderating roles of several coping strategies on the association between pain intensity and pain interference at the daily level using daily diaries in people with CLBP in one design. We used daily diaries to elucidate these dynamics by examining both the person and the day level for three different pain coping strategies, namely, pain rumination, distraction, and reappraisal. Our results build on existing findings concerning the role of pain coping on the pain experience [37] by elucidating the role of *daily* pain appraisals.

As expected, pain rumination, an aspect of pain catastrophizing, emerged as a significant moderator both on the person (i.e., between participants) and on the day (i.e., within participants) level. Specifically, greater rumination increased the associations between pain intensity and pain interference. These results correspond with previous findings [15, 38, 39] and emphasize that pain

Table 1. Participants' demographic characteristics

	Total N = 84 (%)
Sex	
Female	42 (50.0%)
Male	41 (48.8%)
Unknown	1 (1.2%)
Ethnicity	
White	25 (29.8%)
Asian	15 (17.9%)
Black	4 (4.8%)
Other	7 (8.2%)
Unknown	33 (39.3%)
Education	
High school diploma	4 (4.8%)
Some college (no degree)	14 (16.7%)
Associate degree or vocational certificate	5 (5.9%)
Bachelor's degree	29 (34.5%)
Advanced degrees	31 (36.9%)
Unknown	1 (1.2%)
Income	
Less than \$30,000	16 (19.0%)
\$30,000 – \$50,000	9 (10.7%)
\$50,000 - \$70,000	3 (3.6%)
More than \$70,000	47 (56.0%)
Unknown	9 (10.7%)

Table 2. Means, standard deviations, and correlations between study variables (past 24 hours) on the person level across 30 days

Variable	M (range)	SD	1	2	3	4
1. Average pain intensity	45.26 (11–76.46)	16.02				
2. Pain interference	36.31 (0–72.54)	16.96	0.66**			
3. Pain rumination	33.62 (0–72.46)	17.68	0.59**	0.62**		
4. Pain reappraisal	38.25 (0–81.74)	22.69	0.21	0.22*	0.41**	
5. Pain distraction	35.28 (0–100)	23.11	0.58**	0.56**	0.74**	0.62**

Note. M and SD are used to represent mean and standard deviation, respectively.

*Indicates $P < .05$.

**Indicates $P < 0.01$.

Table 3. Person-level moderation analysis predicting pain interference

Predictors	Pain Interference with Daily Activities		
	Estimates	CI	P
Intercept	0.03	−0.14–0.20	.74
Pain intensity	0.29	0.09–0.49	.006
Pain rumination	0.34	0.11–0.57	.004
Pain reappraisal	−0.14	−0.34–0.05	.15
Pain distraction	0.26	−0.02–0.54	.07
Pain intensity X Rumination	0.28	0.07–0.48	.008
Pain intensity X Reappraisal	−0.16	−0.33–0.02	.08
Pain intensity X Distraction	−0.28	−0.52 to − 0.03	.03

CI = Confidence Interval. Significant results appear in bold.

rumination is associated with poorer well-being and functional status. One explanation for this association may be that perseverative focus and attention to the alarming aspects of pain impairs one's ability to attend to other issues and may lead to increased pain perception and negative affect and interfere with adaptive coping strategies [40, 41]. In addition, rumination can be understood using Lazarus and Folkman's (1984) [42] model of appraisal and coping [30]. According to this model, primary appraisal involves attending to and evaluating the threat of a particular event, whereas secondary appraisal involves evaluating the ability to cope with the event. Thus, rumination may correspond to the primary appraisal process involving perseverative focus on pain. Findings from this study highlight the considerable extent to which daily rumination, a primary appraisal process, may exacerbate the association of pain intensity with pain-related interference in daily activities among people with CLBP.

Pain reappraisal also significantly moderated the association between pain and daily functioning at the day level, albeit in a different direction. Specifically, the use of more reappraisal was associated with a lower association between pain intensity and pain interference. This is also aligned with Lazarus and Folkman's (1984) [42] model of appraisal and coping [30], whereby the reappraisal process can be conceptualized as framing the pain in a less negative or harmful way that may lead to the secondary appraisal involving the evaluation of one's ability to apply adaptive cognitive coping strategy.

Adaptive coping, in turn, may decrease the impact of pain intensity and increase self-efficacy to undertake daily activities [43, 44]. Notably, the moderating effect of pain reappraisal between participants (i.e., on the person-level) was smaller and not statistically significant ($P = .08$). This result suggests that reappraisal is beneficial in adjusting the present pain experience, and not necessarily in shaping a global pain interference perception. Future research should focus on the difference between the effects of daily and global pain reappraisal as a coping strategy for pain intensity.

Pain distraction moderated the associations between pain intensity and pain interference, but only at the person level. Importantly, distraction was a significant moderator at the lower levels of pain intensity and distraction, implying that the positive effect of distraction may be limited. Distraction is an intuitive way to attempt to control pain in daily life [45], that is, taking the attention away from the pain to produce analgesia. However, when the pain is intense, distracting from one's pain does not seem to influence the positive associations between pain and pain interference. Although distraction was a significant moderator at the person-level, it was not significant on the day-level. One possible explanation is that for a specific day, for some participants, distraction may be beneficial, while for others or at other times it may not. For a broader discussion of the context dependent nature of the impact of regulation, please see McRae et al. (2011) [46] and English et al. (2017) [47].

The associations between the average pain coping strategies were all positive, even though they played different roles in the associations between pain and pain interference. The association between pain rumination and pain distraction was significantly higher than the association between pain rumination and pain reappraisal. This implies that overall, participants who ruminate over their pain are more likely to also attempt to distract themselves from their pain, than to use a more adaptive pain coping strategy such as pain reappraisal. Future research should focus on the factors that influence the selection of different pain coping strategies, as they would be particularly clinically relevant in formulating programs and intervention for people with CLBP.

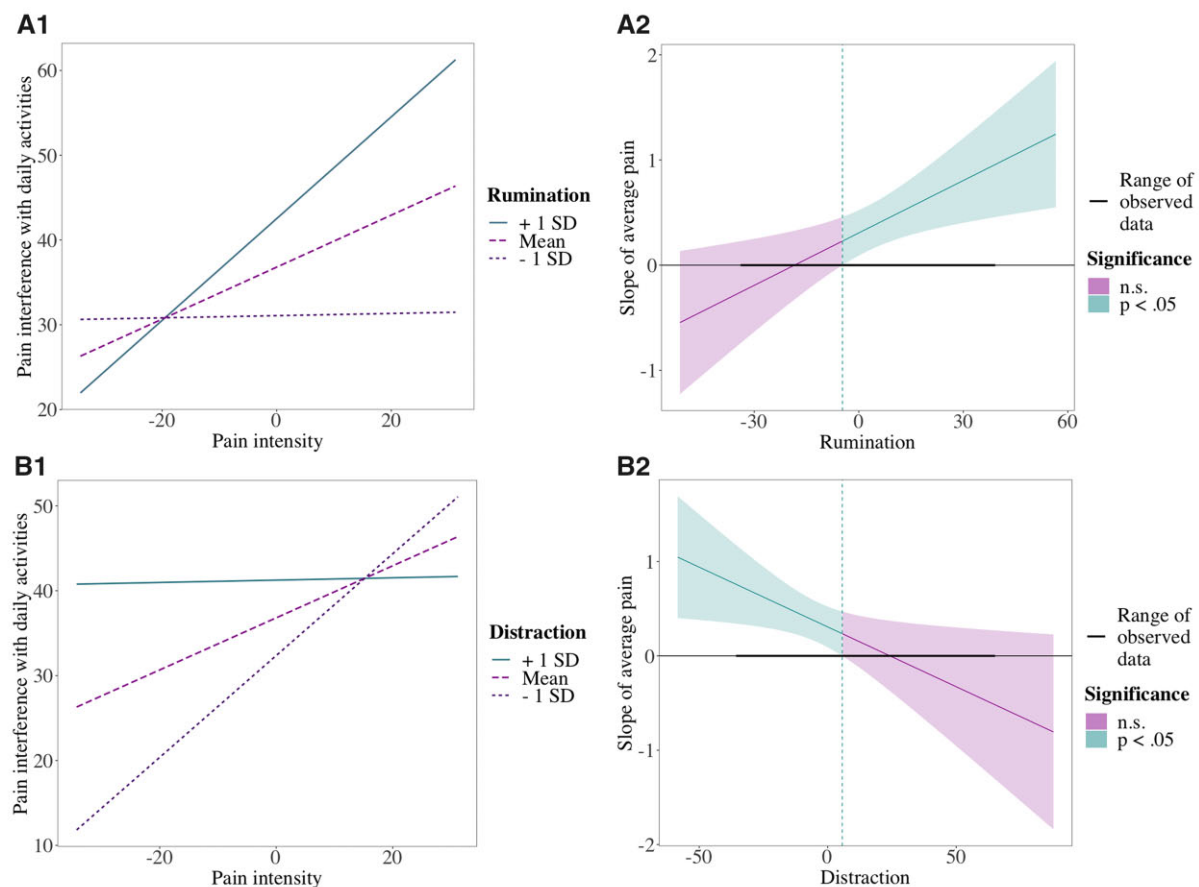


Figure 1. Plot of the moderating roles of rumination (A), and distraction (B) in the association between person-level pain and pain interference. On the left the interaction plot is presented, and on the right the Johnson-Neyman intervals are presented, for significance of predictor slopes.

Table 4. Day-level moderation analysis predicting pain interference

Predictors	Pain Interference with Daily Activities		
	Estimates	CI	P
Intercept	-0.19	-0.31 to -0.06	0.003
Lagged interference	0.004	0.00 - 0.01	<0.001
Pain intensity (P)	-0.13	-0.27 - 0.02	0.080
Pain rumination (P)	0.03	-0.13 - 0.18	0.742
Pain distraction (P)	0.07	-0.14 - 0.28	0.493
Pain reappraisal (P)	-0.02	-0.16 - 0.12	0.760
Pain intensity (D)	0.48	0.42 - 0.55	<0.001
Pain rumination (D)	0.22	0.17 - 0.27	<0.001
Pain distraction (D)	0.15	0.10 - 0.20	<0.001
Pain reappraisal (D)	-0.06	-0.11 to -0.00	0.039
Pain intensity (D) × Rumination (D)	0.06	0.02 - 0.10	0.005
Pain intensity × Distraction(D)	0.02	-0.02 - 0.07	0.371
Pain intensity (D) × Reappraisal (D)	-0.05	-0.09 to -0.01	0.016

P = person level; D = Day level; CI = Confidence Intervals. Significant results are in bold.

Our current study had several strengths. We used daily diaries, a method that is appropriate to study temporal relationships between pain and cognitive factors [20]. The use of daily diaries also enabled us to test both between individual differences as well as within

individual differences. Furthermore, we followed a clinical sample with CLBP for up to 30 days, which is a longer period than most daily diaries [48–50]. Moreover, using three different pain coping strategies in one design enabled us to compare the efficacy of the three strategies.

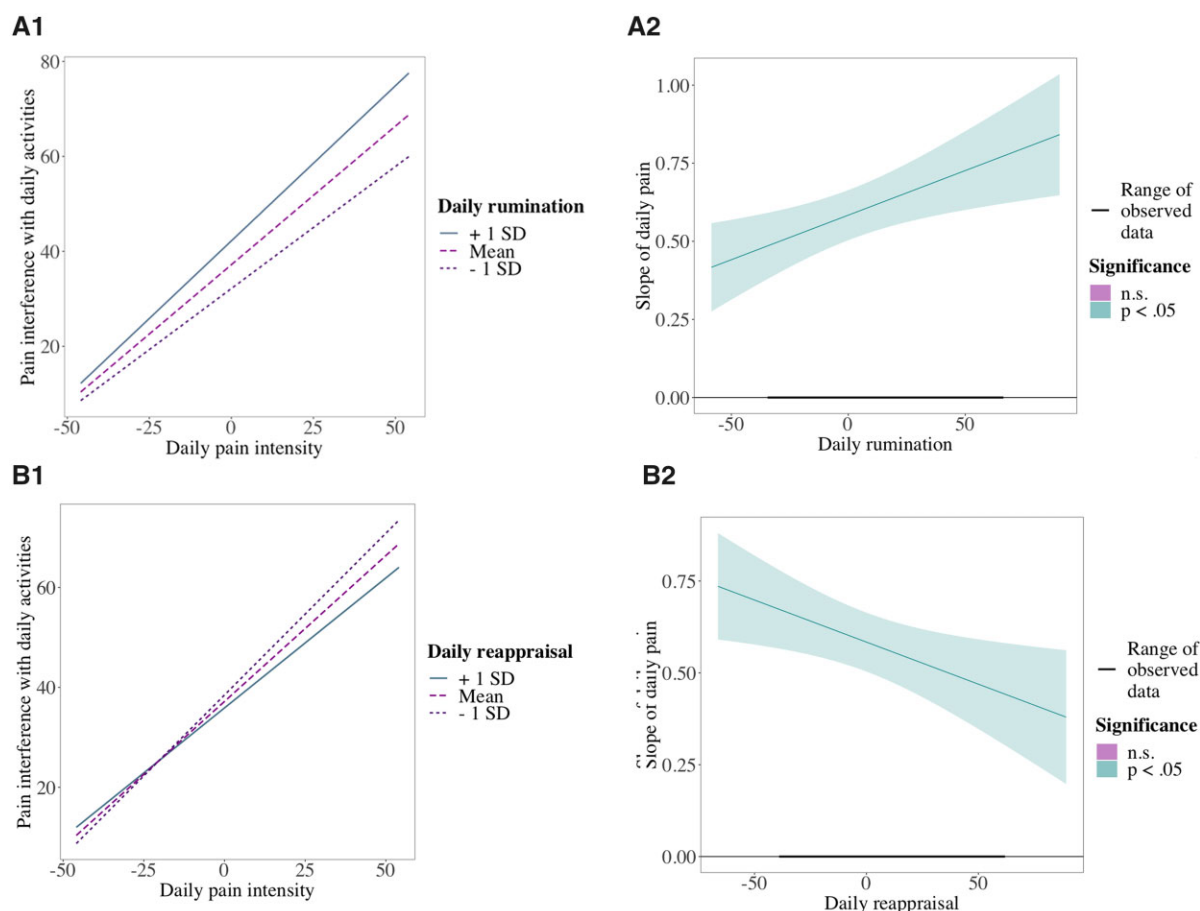


Figure 2. Plot of the moderating roles of Daily rumination (A), and Daily reappraisal (B) in the association between day-level pain and pain interference in daily activities. On the left the interaction plot is presented, and on the right the Johnson-Neyman intervals are presented, for significance of predictor slopes.

Finally, our sample was derived from a CLBP study which carefully attended to the NIH research standards for CLBP.

Limitation and Future Directions

Several limitations should be noted. First, our analyses relied solely on correlations among variables, which hinders causal interpretation. This limitation is somewhat mitigated by the inclusion of lagged pain interference in all models, so that the variance explained by pain intensity and pain coping is above and beyond the variance explained by yesterday's pain interference. Second, each of the variables (i.e., pain intensity, pain interference, and pain coping) was assessed with a single item. This approach was adopted because daily diaries are meant to be short to reduce participant burden and enhance feasibility, especially when are measured for a relatively long time, as was the case in the current study. However, these items were validated by calculating the Pearson correlations between each of these items and relevant and validated emotion regulation and pain catastrophizing questionnaires. We found weak-to-moderate associations between the daily and global assessments, corresponding to previous findings [29]. Third, our sample was younger

and had higher socioeconomic status (higher education and income) than average patients with CLBP, thus limiting generalizability of findings. Future studies should include diverse patient populations. Fourth, we did not control for the time of day the measures were administered. Future studies should control, or at least monitor participants' response, and set it to evening times, when they had a chance to experience the entire day's events. Fifth, we did not have a full account on participants' demographics. The demographics' survey was optional and not mandatory to reduce participants' burden. Therefore, some participants opted not to answer all the questions. Finally, we focused on three coping regulation strategies, which lie in the intersection between affective and pain science. However, it is noteworthy that there are many other important pain coping strategies, such as resting, pacing, seeking social support, exercising, and acceptance. Future work should engage more broadly with other forms of pain coping strategies.

Conclusions

This diary study examined the moderating role of pain coping strategies in the relation between pain and pain

interference with daily activities among individuals with CLBP. Findings suggest that individuals who utilize more reappraisal and distraction when dealing with CLBP may suffer less from CLBP as a challenging health condition. Our findings also highlight the negative impact of rumination on functioning, both at the daily and person-levels, reinforcing the importance of addressing rumination as part of CLBP treatment. Pain coping may be adaptively modified with psychological interventions (e.g., cognitive-behavioral therapy) to help reduce pain and pain-related interference [51]. Future interventions may be tailored to target daily ruminative and reappraisal processes to facilitate adaptive change.

Supplementary Data

Supplementary data are available at *Pain Medicine* online.

References

1. Institute of Medicine (US) Committee on Advancing Pain Research, Care and Education. *Relieving Pain in America: A Blueprint for Transforming Prevention, Care, Education, and Research*. Washington, DC: National Academies Press (US); 2011.
2. Dieleman JL, Cao J, Chapin A, et al. US health care spending by payer and health condition, 1996-2016. *JAMA - J Am Med Assoc* 2020;323(9):863-84.
3. Freburger JK, Holmes GM, Agans RP, et al. The rising prevalence of chronic low back pain. *Arch Intern Med* 2009;169(3):251-8.
4. Dueñas M, Ojeda B, Salazar A, Mico JA, Failde I. A review of chronic pain impact on patients, their social environment and the health care system. *J Pain Res* 2016;9:457-67.
5. Besen E, Gaines B, Linton SJ, Shaw WS. The role of pain catastrophizing as a mediator in the work disability process following acute low back pain. *J Appl Biobehav Res* 2017;22(1):1-13.
6. Sullivan MJL, Thorn B, Haythornthwaite JA, et al. Theoretical perspectives on the relation between catastrophizing and pain. *Clin J Pain* 2001;17(1):52-64.
7. Adamczyk AK, Ligeza TS, Wyczesany M. The dynamics of pain reappraisal: The joint contribution of cognitive change and mental load. *Cogn Affect Behav Neurosci* 2020;20(2):276-93. doi:10.3758/s13415-020-00768-7
8. Goubert L, Crombez G, Eccleston C, Devulder J. Distraction from chronic pain during a pain-inducing activity is associated with greater post-activity pain. *Pain* 2004;110(1-2):220-7.
9. Lin CC. Comparison of the effects of perceived self-efficacy on coping with chronic cancer pain and coping with chronic low back pain. *Clin J Pain* 1998;14(4):303-10.
10. Peres MF, Lucchetti G. Coping strategies in chronic pain | SpringerLink. *Curr Pain Headache Rep* 2010;14(5):331-8.
11. Barry LC, Kerns RD, Guo Z, Duong BD, Iannone LP, Reid MC. Identification of strategies used to cope with chronic pain in older persons receiving primary care from a Veterans Affairs Medical Center. *J Am Geriatr Soc* 2004;52(6):950-6.
12. Reid MC, Barry LC, Kerns RD, Duong BD, Concato J. Coping strategies and their associations with levels of disability or pain, among older veterans receiving primary care. *J Clin Epidemiol* 2002;55(6):629.
13. Buenaver LF, Quartana PJ, Grace EG, et al. Evidence for indirect effects of pain catastrophizing on clinical pain among myofascial temporomandibular disorder participants: The mediating role of sleep disturbance. *Pain* 2012;153(6):1159-66.
14. Sullivan MJL, Sullivan ME, Adams HM. Stage of chronicity and cognitive correlates of pain-related disability. *Cogn Behav Ther* 2002;31(3):111-8.
15. Edwards RR, Cahalan C, Calahan C, Mensing G, Smith M, Haythornthwaite JA. Pain, catastrophizing, and depression in the rheumatic diseases. *Nat Rev Rheumatol* 2011;7(4):216-24.
16. Berman CJ, O'Brien JD, Zenko Z, Ariely D. The limits of cognitive reappraisal: Changing pain valence, but not persistence, during a resistance exercise task. *Int J Environ Res Public Health* 2019;16(19):3739.
17. Affleck G, Urrows S, Tennen H, Higgins P. Daily coping with pain from rheumatoid arthritis: Patterns and correlates. *Pain* 1992;51(2):221-9.
18. Mullen B, Suls J. The effectiveness of attention and rejection as coping styles: A meta-analysis of temporal differences. *J Psychosom Res* 1982;26(1):43-9.
19. Burns JW, Gerhart J, Rizvydeen M, Kimura M, Burgess HJ. Morning bright light treatment for chronic low back pain: Potential impact on the volatility of pain, mood, function, and sleep. *Pain Med (United States)* 2020;21(6):1153-61.
20. May M, Junghaenel DU, Ono M, Stone AA, Schneider S. Ecological momentary assessment methodology in chronic pain research: A systematic review. *J Pain* 2018;19(7):699-716.
21. Jensen MP, Turner JA, Romano JM, Strom SE. The chronic pain coping inventory: Development and preliminary validation. *Pain* 1995;60(2):203-16.
22. Gendreau M, Hufford MR, Stone AA. Measuring clinical pain in chronic widespread pain: Selected methodological issues. *Best Pract Res Clin Rheumatol* 2003;17(4):575-92.
23. Jamison RN, Sbrocco T, Parris WCV. The influence of physical and psychosocial factors on accuracy of memory for pain in chronic pain patients. *Pain* 1989;37(3):289-94.
24. Redelmeier DA, Kahneman D. Patients' memories of painful medical treatments: Real-time and retrospective evaluations of two minimally invasive procedures. *Pain* 1996;66(1):3-8.
25. Stone AA, Broderick JE, Shiffman SS, Schwartz JE. Understanding recall of weekly pain from a momentary assessment perspective: Absolute agreement, between- and within-person consistency, and judged change in weekly pain. *Pain* 2004;107(1-2):61-9.
26. Schneider S, Stone AA. Ambulatory and diary methods can facilitate the measurement of patient-reported outcomes. *Qual Life Res* 2016;25(3):497-506.
27. Shiffman S, Stone AA, Hufford MR. Ecological momentary assessment. *Annu Rev Clin Psychol* 2008;4:1-32.
28. Deyo RA, Dworkin SF, Amtmann D, et al. Report of the NIH task force on research standards for chronic low back pain. *Int J Ther Massage Bodyw Res Educ Pract* 2015;8(3):16-33.
29. Koval P, Kalokerinos EK, Greenaway KH, et al. Emotion regulation in everyday life: Mapping global self-reports to daily processes. *Emotion* 2022. doi:10.1037/emo0001097.
30. Sullivan MJL, Bishop SR, Pivik J. The pain catastrophizing scale: Development and validation. *Psychol Assess* 1995;7(4):524-32.
31. Weiss NH, Bold KW, Sullivan TP, Armeli S, Tennen H. Testing bidirectional associations among emotion regulation strategies and substance use: A daily diary study. *Addiction* 2017;112(4):695-704.
32. Gross JJ, John OP. Individual differences in two emotion regulation processes: Implications for affect, relationships, and well-being. *J Pers Soc Psychol* 2003;85(2):348-62.

33. Nolen-Hoeksema S. Responses to depression and their effects on the duration of depressive episodes. *J Abnorm Psychol* 1991;100(4):569–82.
34. Pinheiro J, Bates D, DebRoy S, Sarkar D, Team RC. Linear and nonlinear mixed effects models. *R Packag Version* 2007;3(57):1–89.
35. Bolger N, Laurenceau JP. *Intensive Longitudinal Methods: An Introduction to Diary and Experience Sampling Research*. New York: Guilford Press; 2013.
36. Preacher KJ, Curran PJ, Bauer DJ. Computational tools for probing interactions in multiple linear regression, multilevel modeling, and latent curve analysis. *J Educ Behav Stat* 2006;31(4):437–48.
37. Tracey I. Getting the pain you expect: Mechanisms of placebo, nocebo and reappraisal effects in humans. *Nat Med* 2010;16(11):1277–83.
38. Widerstrom-Noga EG, Felipe-Cuervo E, Yezierski RP. Chronic pain after spinal injury: Interference with sleep and daily activities. *Arch Phys Med Rehabil* 2001;82(11):1571–7.
39. Meints SM, Stout M, Abplanalp S, Hirsh AT. Pain-related rumination, but not magnification or helplessness, mediates race and sex differences in experimental pain. *J Pain* 2017;18(3):332–9.
40. Bartley EJ, Rhudy JL. The influence of pain catastrophizing on experimentally induced emotion and emotional modulation of nociception. *J Pain* 2008;9(5):388–96.
41. Sullivan MJL, Stanish W, Waite H, Sullivan M, Tripp DA. Catastrophizing, pain, and disability in patients with soft-tissue injuries. *Pain* 1998;77(3):253–60.
42. Lazarus RS, Folkman S. *Stress, Appraisal, and Coping*. New York: Springer publishing company; 1984.
43. Touche RL, Grande-Alonso M, Arnés-Prieto P, Paris-Alemany A. How does self-efficacy influence pain perception, postural stability and range of motion in individuals with chronic low back pain? *Pain Physician* 2019;1(22;1):E1–E13.
44. Turner JA, Anderson ML, Balderson BH, Cook AJ, Sherman KJ, Cherkin DC. Mindfulness-based stress reduction and cognitive behavioral therapy for chronic low back pain: Similar effects on mindfulness, catastrophizing, self-efficacy, and acceptance in a randomized controlled trial. *Pain* 2016;157(11):2434–44.
45. Leventhal H. I know distraction works even though it doesn't! *Heal Psychol* 1992;11(4):208–9.
46. McRae K, Heller SM, John OP, Gross JJ. Context-dependent emotion regulation: suppression and reappraisal at the burning man festival. *Basic Appl Soc Psych* 2011;33(4):346–50.
47. English T, Lee IA, John OP, Gross JJ. Emotion regulation strategy selection in daily life: The role of social context and goals. *Motiv Emot* 2017;41(2):230–42.
48. Middendorp H, Lumley MA, Moerbeek M, Jacobs JWJ, Bijlsma JWJ, Geenen R. Effects of anger and anger regulation styles on pain in daily life of women with fibromyalgia: A diary study. *Eur J Pain* 2010;14(2):176–82.
49. Burns JW, Gerhart JI, Bruehl S, et al. Anger arousal and behavioral anger regulation in everyday life among patients with chronic low back pain: Relationships to patient pain and function. *Heal Psychol* 2015;34(5):547–55.
50. Stone AA, Broderick JE, Schwartz JE, Shiffman S, Litcher-Kelly L, Calvanese P. Intensive momentary reporting of pain with an electronic diary: Reactivity, compliance, and patient satisfaction. *Pain* 2003;104(1-2):343–51.
51. Hoffman BM, Papas RK, Chatkoff DK, Kerns RD. Meta-analysis of psychological interventions for chronic low back pain. *Heal Psychol* 2007;26(1):1–9.